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# Machine learning for predicting landslide risk of Rohingya refugee camp infrastructure

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**ABSTRACT**

Since the dawn of human civilization, forced migration scenarios have been witnessed in different regions and populations, and is still present in the twenty-first century. The current largest population of stateless refugees in the world, the Rohingya people, reside in the southeastern border region of Bangladesh. Due to rapid expansion of refugee camps and lack of suitable locations, a large proportion of the infrastructure are at risk of landslides. This study aims to use machine learning for predicting landslide risk of camp infrastructure using geospatial features. Four supervised classification algorithms have been employed viz., (i) Logistic Regression (LR), (ii) Multi-Layer Perceptron (MLP), (iii) Gradient Boosted Trees (GBT) and (iv) Random Forest (RF) and applied on preprocessing varied versions of features. Results show that RF achieves accuracy of 76.19% and AUC of 0.76 on un-scaled features which is higher than all other algorithms. The applications of the study reside in refugee management and landslide susceptibility mapping of Rohingya camps, which can both potentially save refugee lives and serve as a case study for global applications.

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Machine learning; risk prediction; Rohingya; landslide; geospatial feature

## 1. Introduction

Refugee flows, falling under the umbrella term of forced migration, have existed since the beginning of civilization, triggered by war, famine, crisis, differences in ethnicity and race, etc. (Castles, 2003). However, it is still present in today's world (Massey, 2003). Currently, the largest population of stateless refugees in the world, the Rohingya people, reside in Bangladesh (UNHCR, 2019). The root cause of this mass refugee flow is linked to the Myanmar government not recognizing the Rohingya ethnicity as lawful citizens (Lewa, 2009). Since 1978, the persecution of the population (Kipgen, 2014) led to the eventual refugee flow into neighbouring Bangladesh, resulting in decades of refugee movement into the host country (Ullah, 2011; Zarni & Cowley, 2014).

There are several negative implications of refugee movement as observed within the Rohingya population. For example, studies on Rohingya refugees have focused on communicable epidemic diseases (Alam et al., 2019), mental health (Khan et al.,

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2019; Majeed, 2019; Shaw, Pillai, & Ward, 2019), outbreak of diphtheria (Rahman & Islam, 2019), maternal health services (Khan & DeYoung, 2019), vaccination delivery (Jalloh et al., 2019), etc. Consequently, health facilities for Rohingya population in Bangladesh are of utmost importance for their well-being. The need for educational facilities, i.e. learning centres are also necessary for refugee populations (Letchamanan, 2013; Lewa, 2009; Ullah, 2011). On the other hand, sanitation systems in the Rohingya camps are interrelated to disease and health risk (Mahmood, Wroe, Fuller, & Leaning, 2017) where hand pump tube wells and latrines play an important role in management and minimization of health risk. Refugee management entities have invested sufficient funds as well as taken necessary steps for establishing infrastructure, i.e. general infrastructure, health facilities, learning centres, tube-wells and latrines.

Due to rapid expansion of refugee camps, land was cut and altered without stability being considered as significant priority. As a result, a large proportion of the camp infrastructure is at risk of landslides. Since the mass arrivals of Rohingya refugees began on 25 August 2017 (BBC, 2017), the first monsoon was faced by approximately a million refugees for the first time in the year 2018, which was a significant issue for the management authorities (Cousins, 2018) especially since the refugees were unprepared for the monsoon season and landslides (BBC, 2018). Temporary nature of camp infrastructure has led to landslide risk (Beaubien, 2018; IOM, 2018) and evacuation (Hussein & Hossain Shakil, 2018) as well as eventual landslide incidents (Project, 2018; Watch, 2018) and death of refugees (Petley, 2018). Natural disasters such as landslides can wreak havoc on public health conditions, specially for refugees who are more susceptible to health risks (Ahmed, Orcutt et al., 2018). Landslide risk refers to the probability of the occurrence of landslides for an entity (such as infrastructure) in a specific geographic location (dependent on causal variables, i.e. geospatial features) multiplied by the vulnerability (Fell, 1994). The large number of at-risk infrastructure combined with the consequences of landslides recommends the need for the employment of scientific methods for predicting the landslide risk of Rohingya camps.

Currently, there is no existent study focusing on a machine learning based approach of landslide risk prediction for Rohingya refugee camp infrastructure. Machine learning models for supervised binary classification can be used for predicting the labels of samples based on corresponding features. In the context of landslide risk prediction, the models can be employed for predicting whether an infrastructure location is prone to landslide risk or not based on geospatial features. The objectives of this study are to (i) analyse the geospatial profile of the study area in southeastern border region of Bangladesh where most Rohingya camps are located, (ii) assess the landslide risk of refugee camp infrastructure and (iii) apply machine learning algorithms in an attempt to predict landslide risk of camp infrastructure from geospatial features. The models developed in this study can forecast if a particular potential point in a map, i.e. a physical area where actual humans live, is prone to landslide and with what kind of certainty, having the opportunity to contributing to directly save lives. Furthermore, this study is not only the first machine learning driven approach to landslide risk prediction of Rohingya camps, it is in fact the very first study in Bangladesh which utilizes machine learning based methodology for landslide risk prediction.

## 2. Related work

Landslide risk prediction involves the pre-adjustment of a model (training) to predict the risk of a location from features. The value of vulnerability lies between 0 (no damage) and 1 (maximum damage). For this study, the vulnerability is considered as maximum damage for mainly two reasons viz., (i) the camp infrastructure is not permanent as there is ongoing political debate about the final residence of the Rohingya refugees and (ii) relevant information about infrastructure such as construction materials, dimension and so on of the infrastructure are not available. Before the arrival of wide-reaching machine learning models, landslide risk prediction was achieved using field survey and photointerpretation (Aniya, 1985), followed by probabilistic models (Bernknopf, Campbell, Brookshire, & Shapiro, 1988), weighted sum approach (Nagarajan, Roy, Kumar, Mukherjee, & Khire, 2000) and image processing techniques (Gökceoglu & Aksoy, 1996). Machine learning models have been shown to outperform traditional models such as the Multi-Criteria Decision-Making (MCDM) analysis (Khosravi et al., 2019) for natural disaster prediction. Hybrid models in the form of ensembles of base classifiers have also been used for the task (Thai Pham et al., 2018). Landslide susceptibility of areas in the southeastern region of Bangladesh have been studied (Ahmed, 2015a, 2015b; Ahmed & Dewan, 2017) but none employed any machine learning and/or data mining based methodology nor did any study focus on landslide risk prediction of Rohingya camps.

The very first machine learning algorithm (in the form of statistical models) frequently being used to map landslides was Logistic Regression (LR) (Dai, Lee, Li, & Xu, 2001; Lee & Min, 2001) and the trend continues (Ayalew & Yamagishi, 2005; Lee, 2005; Lee & Pradhan, 2007; Lee & Sambath, 2006; Ohlmacher & Davis, 2003; Van Den Eeckhaut et al., 2006). Then came the artificial neural network models (Pradhan & Lee, 2010a, 2010b; Yesilnacar & Topal, 2005; Yilmaz, 2009). In most cases, back-propagated Multi-Layer Perceptron (MLP) models were used. The MLP models were directly compared with the previously ubiquitous LR models. Studies applying both LR and MLP found that results provided by the MLP were better (Pradhan & Lee, 2010b) and more realistic (Yesilnacar & Topal, 2005). Another study comparing both models found that area under curve (AUC) value of neural network model (0.852) was higher than that of LR (0.842) (Yilmaz, 2009). However, works have also shown that LR model can outperform MLP model. For example, a study found that LR model achieved accuracy of 89.59% whereas Artificial Neural Network (ANN), i.e. MLP model achieved 83.55% (Pradhan & Lee, 2010a). Afterwards, bagging and boosting methods, i.e. Gradient Boosted Trees (GBT) (Kim, Lee, Jung, & Lee, 2018; Lombardo, Cama, Conoscenti, Märker, & Rotigliano, 2015; Vorpahl, Elsenbeer, Märker, & Schröder, 2012; Youssef, Pourghasemi, Pourtaghi, & Al-Katheeri, 2016) and Random Forest (RF) (Belgiu & Drăguţ, 2016; Catani, Lagomarsino, Segoni, & Tofani, 2013; Chen, Li, Wang, Chen, & Liu, 2014; Chen et al., 2018, 2017; Hong, Pourghasemi, & Pourtaghi, 2016; Stumpf & Kerle, 2011) were applied for the task. The computation heavy models that were previously unfeasible due to hardware constraints became then possible. A recent study comparing GBT and RF found that GBT achieved higher accuracy, 84.87% for regression and 85.98% for classification whereas RF achieved 79.34% for regression and 79.18% for classification (Kim et al., 2018). The four aforementioned machine learning algorithms viz., (i) LR, (ii) MLP, (iii) GBT and (iv) RF have been employed in this study.

Landslide prediction requires the inclusion of determinant geospatial features which must be selected carefully prior to machine learning. The geospatial features/variables considered in this study are influenced by the aforementioned studies, emphasizing on a seminal work which used machine learning for landslide risk prediction (Marjanović, Kovačević, Bajat, & Voženílek, 2011) (for details refer to Section 3.2.1 and 3.3) to compare ranking of features. Practically, all of the mentioned studies have used the same geospatial features. For example, elevation, Normalized Difference Vegetation Index (NDVI), Topographic Wetness Index (TWI) and lithology are used by almost every single study. Depending on the location of the study area and availability of datasets, studies tend to incorporate more features with higher spatial resolution and/or accuracy/dependability. However, for Bangladesh there is both a limited number of available geospatial features as well as constrained resolution/dependability. For example, most fast growing first-world countries use elevation data retrieved by state employed services. Whereas, there is no such analogous and specialized dataset available for Bangladesh. Thus the global elevation raster dataset is used which is of worse spatial resolution than that of national scale datasets. A similar issue can be observed for the landslide risk dataset (target variable). Almost all discussed studies had access to spatial landslide risk data, i.e. the landslide risk of *regions* whereas the only landslide risk data available for all Rohingya camp locations are in the form of *points*/coordinates. However, the basic principles remain the same in the context of use of geospatial data for predicting landslide risk.

## 3. Materials and methods

### 3.1. Study area

The study area lies between latitude 20°53′46.7″N and 21°14′29.8″N, and longitude 92°02′08.2″E and 92°18′27.0″E, excluding surrounding water bodies, such as the Bay of Bengal to the west and Naf river to the east. Myanmar is also excluded from the study area since the study focuses on infrastructures located in Bangladesh, exclusively shown in Figure 1. The study area lies within the Cox's Bazar *district* (second largest administrative unit after *division*) of Bangladesh, which includes the Teknaf Game Reserve. The Rohingya refugee camps are situated in Teknaf and Ukhia *upazila* (third largest administrative unit). *Unions* (fourth largest administrative unit) in Teknaf containing refugee camps are Whykong, Nihila, Baharchhara and Sabrang. Unions with refugee population in Ukhia upazila are Palong Khali, Raja Palong, Haldia Palong, Jalia Palong and Ratna Palong. This region has witnessed higher number of occurrences of landslides compared to other regions of Bangladesh, leading to aforementioned research emphasizing on the landslide risk assessment of the region in the past.

### 3.2. Data

#### 3.2.1. Geospatial feature data

The four available geospatial variables have following information-gain (IG) ranking according to Marjanović et al. (2011) – elevation (rank 1), lithology (rank 2), NDVI (rank 3) and TWI (rank 4). However, a different scenario is observed for the current study as

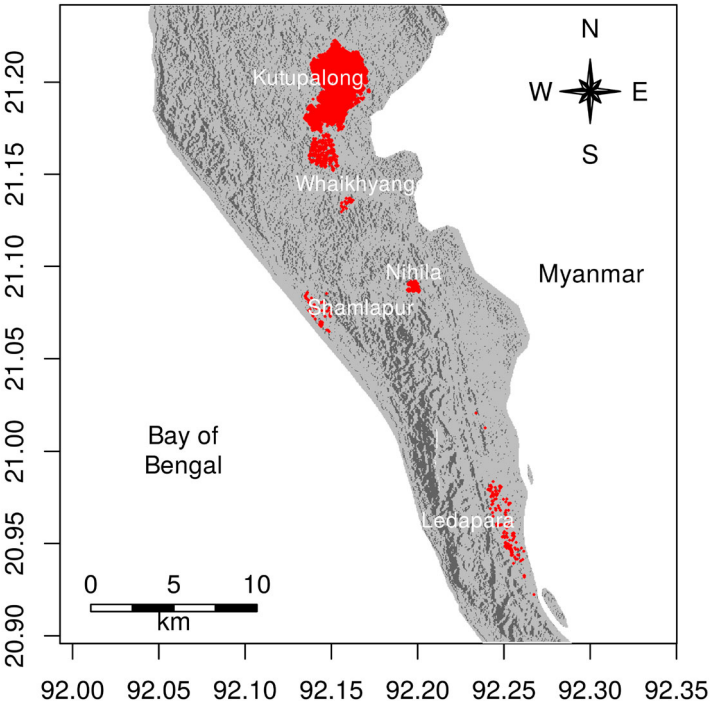



**Figure 1.** Hillshade of the study area in the southeastern border region of Bangladesh; the red points represent the locations of camp infrastructure.

seen in Table 1. Elevation dataset is extracted from the gap-filled global 30 m Digital Elevation Model (DEM) (Farr et al., 2007), lithology dataset is extracted from the Surface Geology Map of Bangladesh (USGS, 2001), NDVI is calculated from Landsat 8 satellite image (USGS, 2018), and TWI is computed using global flow accumulation (Lehner, Verdin, & Jarvis, 2008) and slope, computed from DEM (Farr et al., 2007) datasets. Geospatial features and respective methods for deriving sampling-ready datasets are discussed in Sections 3.3 and 3.4. The information gain (IG) (Kent, 1983) and feature importance (FI) (Porter, Bareiss, & Holte, 1990; Zien, Krämer, Sonnenburg, & Rätsch, 2009) ranks of utilized geospatial features are also shown in Table 1.

### *3.2.2. Landslide risk data*

The general infrastructure (Table 2) consists of mosques, child friendly spaces, madrassas, women friendly spaces, distribution centres, field offices and community centres. Health facilities provide medical attention to the refugees residing in the camps. The learning centres provide educational facilities to refugee population. Hand pump tube wells are

**Table 1.** Source, resolution and ranking of geospatial features.

| Feature | IG rank | FI rank | Resolution | Dataset | Source |
|---|---|---|---|---|---|
| Elevation | 4 | 3 | 30 m | DEM | Farr et al. (2007) |
| Lithology | 3 | 4 | Not applicable | Surface geology | USGS (2001) |
| NDVI | 1 | 1 | 30 m | Landsat-8 satellite image | USGS (2018) |
| TWI | 2 | 2 | 30 m | Flow accumulation | Lehner et al. (2008) |

**Table 2.** Source and frequency of refugee camp infrastructure.

| Type | Source | Number |
|---|---|---|
| General infrastructure | REACH Initiative (REACH, 2018a, 2018c) | 1365 |
| Health facilities | REACH Initiative (REACH, 2018a) | 168 |
| Learning centres | REACH Initiative (REACH, 2018a) | 849 |
| Hand pump tube wells | REACH Initiative (REACH, 2018b) | 5945 |
| Latrines | REACH Initiative (REACH, 2018b) | 9348 |
| Total | | 17,675 |

installed to access groundwater through mechanical hand driven pumps. Latrines play a major role in the sanitary health of the refugee camp residents. The data is derived from Round 4 infrastructure mapping performed by the REACH Initiative. Landslide analysis was conducted by Asian Disaster Preparedness Center (ADPC) and United Nations High Commissioner for Refugees (UNHCR). Landslide risk of each of the 17,675 recorded infrastructure is denoted by a corresponding value of either *Yes* for landslide risk presence or *No* for landslide risk absence. The Round 4 infrastructure mapping was performed to primarily assess the needs and conditions (including landslide susceptibility/risk) of camp infrastructure.

## 3.3. Geospatial features

### 3.3.1. Elevation

The feature with the highest information-gain rank, i.e. elevation, is obtained from the global DEM raster dataset. Each cell (raster grid cell of dimensions 30 m × 30 m) has an associated value for land elevation above the sea level in metres (m). Thus the side length of each raster pixel represents 30 m in physical reality. Virtually, every aforementioned study on landslide susceptibility mapping incorporates the elevation of the terrain into the models. Other features, derived from the elevation dataset, are needed for calculating other geospatial features. For example, slope in radians is required for the calculation of TWI. Since slope is calculated solely from elevation, it is highly correlated to elevation itself. This explains why the results remain unaltered when including slope as a feature. Consequently, only elevation is included as a feature while slope is excluded and only used only for the calculation of TWI.

### 3.3.2. Lithology

Lithology data is collected from the Bangladesh Surface Geology geospatial map. The study area is divided into regions with specific geological characteristics and nomenclature, i.e. Girujan Clay (from Pleistocene and Neogene), Tipam Sandstone (Neogene), etc. Thus lithology is a nominal (and categorical) feature. The data is retrieved in shapefile format, compared to continuous and numeric features (elevation, NDVI and TWI) which are retrieved in raster format.

### 3.3.3. NDVI

The Normalized Difference Vegetation Index (NDVI) functions on the principle that regions with live vegetation absorb radiation from the red wavelength of the electromagnetic spectrum and reflect radiation from the near-infrared wavelength of the electromagnetic spectrum. Mathematically, NDVI is the ratio of the difference between and sum of TIR and

Red band value as shown in Equation (1). The value for NDVI always ranges between −1 and +1 inclusive. For example, values close to −1 indicate that the location is water and values close to +1 indicates the presence of dense green leaves. Values close to 0 mean there are no vegetation. Majority of the study area has NDVI between 0.5 and 0.75.

$$NDVI = \frac{NIR - RED}{NIR + RED}. \tag{1}$$

#### 3.3.4. TWI

The Topographic Wetness Index (TWI) (Sörensen, Zinko, & Seibert, 2006) is frequently used as a feature for landslide risk prediction. While greater value of TWI indicates drainage depressions, smaller value indicates crests and ridges. The formula for TWI is as shown in Equation (2).

$$TWI = \ln\left(\frac{A}{\tan\beta}\right), \tag{2}$$

$$A = (F + 1) \times s, \tag{3}$$

where the upstream contributing area is denoted by $A$ (in $m^2$) and $\beta$ is the terrain slope raster derived from DEM dataset. The upstream contributing area $A$ is calculated by the flow accumulation, $F$ (Lehner et al., 2008) shown in Equation (3). The cell size $s$ is the spatial resolution of the flow accumulation raster $F$, i.e. 15 arc-seconds (Lehner et al., 2008).

### 3.4. Preprocessing

The Shuttle Radar Topography Mission (STRM) (Farr et al., 2007) provides global scale land elevation raster dataset, the Digital Elevation Model (DEM), having spatial resolution of 30 m. The dataset is available as gap filled and is then interpolated at 30 m spatial resolution. Elevation data is extracted from this dataset. Slope (which is required for the calculation of TWI) is also derived from DEM. Mathematical operation between two rasters of different spatial resolutions (30 m for slope and 15 arc seconds for flow accumulation) results in output raster having the lower of the spatial resolutions of the two input rasters, i.e. 30 m. While considering the formula for TWI, the natural logarithm function ($\log_e$) as well as all logarithms are undefined in the negative input space. Consequently, there are some holes in the computed TWI raster due to the input of the logarithm function, i.e. $A/\tan\beta$ being negative which are filled using nearest neighbour approach. The NDVI is calculated from Landsat 8 2018 Tier 1 Collection 1 Top Of Atmosphere (TOA) reflectance satellite image. All rasters, i.e. elevation, NDVI and TWI are clipped according to extent of the study area and exported. After all four features have been preprocessed for sampling, i.e. providing the feature values of the locations where infrastructure exists, values at locations of the 17,675 infrastructure locations are extracted. The data type of the feature Lithology was string which was first integer encoded, then one hot encoded. The target column is also initially recorded as string data type (*Yes* for landslide risk presence and *No* for landslide risk absence) which is binary encoded, resulting in class 0 for landslide risk absence and class 1 for landslide risk presence. The units for the four geospatial features used in this study are quite different from each other. NDVI is in metres (m), TWI and NDVI are both

unit-less but with different ranges (discussed in upcoming sections) and lithology is nominal having categories rather than units. Thus the resultant data set is scaled using three methods viz., (i) min–max scaling (Equation 4), (ii) standardization (Equation 5) and (iii) normalization (Equation 6) providing four (including the un-scaled dataset) scale varied versions of the data set to assess the effect of scaling method on the features. $x$ is the un-scaled sample from the feature. The minimum, maximum, mean and standard deviation of the feature are $x_{\min}$, $x_{\max}$, $x_{\mu}$ and $x_{\sigma}$ respectively. Another version of the dataset is created by filtering to remove 10% of outliers from training data using the Isolation Forest algorithm (Liu, Ting, & Zhou, 2008). The testing data is kept the same for all configurations to ensure absence of bias.

$$x_{\min-\max\,\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{4}$$

$$x_{\text{standardized}} = \frac{x - x_{\mu}}{x_{\sigma}} \tag{5}$$

$$x_{\text{normalized}} = \frac{x - x_{\mu}}{x_{\max} - x_{\min}} \tag{6}$$

## 3.5. Machine learning algorithms

### 3.5.1. Logistic regression

The logistic regression model (Hosmer, Lemeshow, & Sturdivant, 2013) operates in two major steps. First, the weighted sum $z$ (where the weights are obtained after model training) of the inputs (the bias term can be thought of input 1 where the bias term is equal to the weight) is calculated (Equation 7). Every feature $x_i$ has a corresponding weight $w_i$.

$$z = \sum_{i=0}^{n} w_i x_i \tag{7}$$

It is then used as input of the logistic sigmoid activation function (Equation 8). The activation function $\phi(z)$ takes the previously calculated weighted sum $z$ as input. The function squashes the input to provide a value between 0 and 1, which can be thought of as the probability of being in a certain class. In most cases, the decision boundary is at 0.5. Given the binary classification problem in this study, an output of 0 or close to 0 signifies the respective infrastructure location having less risk of landslides and an output of 1 or close to 1 signifies the respective infrastructure location having high risk of landslides depending on the decision boundary. The 'logistic' in Logistic Regression is due to the use of the logistic sigmoid activation function (Equation 8).

$$\phi(z) = \frac{1}{1 + e^{-z}} \tag{8}$$

### 3.5.2. Multi-layer perceptron

The multilayer perceptron model (Rosenblatt, 1958) has layer(s) of artificial neurons (basic units of neural network). Equation (9) shows that the activation of neuron $i$ in layer $j$, denoted by $a_i^j$, is dependent on an activation function $f$ (which could be

sigmoid, hyperbolic tangent, arc tangent, identity, rectified linear, etc.) which takes the weighted sum of the activations of the previous layer. The summation of product of activation of neuron $k$ in layer $(j-1)$, denoted by $a_k^{j-1}$ and weight $k$ of layer $j$, denoted by $w_k^j$. The initial inputs of the models can be thought of as the first (non-hidden) layer. Since lithology is a nominal feature it is one-hot encoded. Nominal variables, i.e. categorical features can be in certain discrete states. In machine learning, nominal variables are generally one-hot encoded as a part of preprocessing. The number of resultant one-hot encoded features depends on the number of unique states. There are six lithology classes for the study area (refer to Table 3 which results in the conversion of lithology feature into 6 binary features (one for each of the lithology classes). When an infrastructure falls within a specific lithology class, the corresponding binary feature is encoded as 1 and the other 5 binary features are encoded as 0. Considering the three numeric features (i.e. NDVI, TWI and elevation) and the six features for lithology, there are nine inputs to the MLP and nine corresponding neurons in the first layer. The hidden layer used has nine neurons, identical to the number of inputs. The output layer (final layer) has a single layer, which either outputs 1 for landslide risk presence or 0 for landslide risk absence as shown in Figure 2. The hidden layer(s) in MLP serve the purpose of extracting intermediate higher level features which would otherwise not be available without the use of hidden layer(s). In general, higher number of hidden layers translate to a model which can learn very complex rules, aiding in improvement of accuracy. However, considering the four geospatial features used in this study, a single hidden layer is sufficient for the task of landslide risk prediction and same number of layer has also been used in the existing literature (Pradhan & Lee, 2010a).

$$d_i^j = f\left(\sum_{k=0}^{n} w_k^j a_k^{j-1}\right). \tag{9}$$

### 3.5.3. Gradient boosted trees

Gradient Boosted Trees refer to Gradient Boosting (Friedman, 2001, 2002) applied on Decision Trees (Breiman, 2017). It is an ensemble method where $n$ trees are constructed then. Individual decision tree-based learners are trained and the output of the model, $\hat{F}(x)$, is the weighted sum of the output of the weak learners plus a constant (Equation 10). The weak learners are trained sequentially (boosted), i.e. one after another; the error from previous weak learners is incorporated into training adjustment for subsequent weak learner. However, for binary classification, which is the case for this study, a single

**Table 3.** Lithology category abbreviations and corresponding landslide risk summary.

| Abbreviation | Full form | Landslide risk absent | Landslide risk present | Risk ratio |
|---|---|---|---|---|
| QTg | Girujan Clay (Pleistocene and Neogene) | 6069 | 1986 | 0.3272 |
| QTdt | Dupi Tila Formation (Pleistocene and Pliocene) | 2902 | 1578 | 0.5438 |
| Tt | Tipam Sandstone (Neogene) | 1841 | 832 | 0.4519 |
| H20 | Wide river | 2051 | 294 | 0.1433 |
| Tbb | Boka Bil Formation (Neogene) | 102 | 2 | 0.0196 |
| Csd | Beach and dune sand | 18 | 0 | 0.0000 |

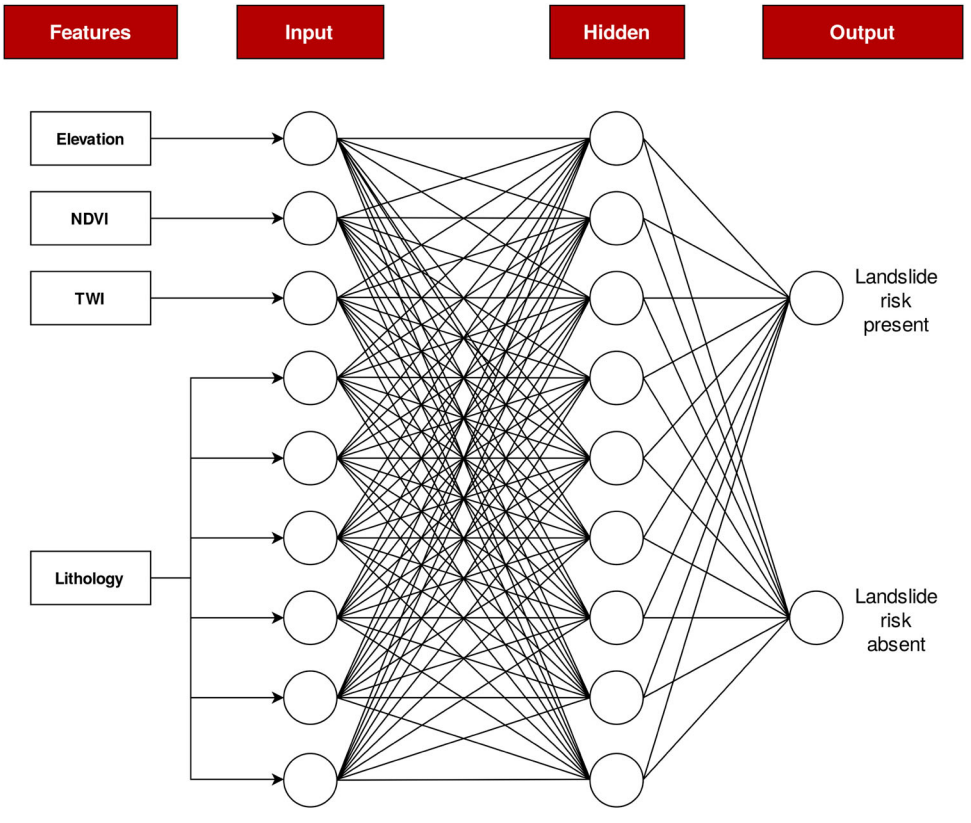



**Figure 2.** Architecture of the Multi-Layer Perceptron used in this study.

tree is created. For the $m$th weak learner ($1 \leq m \leq M$), there is a corresponding weight, $\gamma_m$ which is multiplied with the approximator function $h_m(x)$ which are then summed and added to a constant $c$. The weak learners optimally partition the feature space in order to segregate the samples minimizing misclassifications. The ensemble determines the final prediction based on the votes and the weights of learners (which indicate the importance of a weak learner).

$$\hat{F}(x) = \sum_{m=1}^{M} \gamma_m h_m(x) + c. \tag{10}$$

### *3.5.4. Random forest*

Random Forest (Breiman, 2001) is also an ensemble model, where the prediction is an aggregation of the outputs from the individual decision trees. The algorithm has been used for land cover classification of the southeastern border region of Bangladesh (Ahmed, Islam, Hasan, Motahar & Sujauddin, 2019; Hassan, Smith, Walker, Rahman, & Southworth, 2018). As shown in Equation (11), the outputs of the individual weak learners are averaged. For the $b$th weak learner ($1 \leq b \leq B$), there is a corresponding function $f_b(x)$. The algorithm uses bootstrap aggregating (bagging) where the weak learners are trained in parallel, i.e. simultaneously. Unlike GBT, the algorithm does not have

weighted weak learners, the mean output of B learners is used to determine the class of a specific sample.

$$\hat{F}(x) = \frac{1}{B}\sum_{b=1}^{B} f_b(x). \tag{11}$$

### 3.6. Description of analysed cases

The multi-collinearity of the geospatial features is analysed using the variance inflation factor (VIF) which has been used for this task previously (Dou et al., 2019). A VIF value larger than 10 indicates multi-collinearity. The un-scaled version of the data set as well as the min-max scaled, standardized and normalized versions is used for comparison in training accuracy and testing accuracy of the four aforementioned machine learning models. Each of the four differently scaled version of the datasets is then trained on and tested independently. 80% of the dataset is used for training and 20% is used for testing. Even though the dataset is split randomly, the same randomness is maintained throughout experiments using different algorithms and scale varied versions of features, i.e. though the training and testing data samples are chosen randomly, the same training and testing samples were used for the 4 algorithms on the 5 preprocessing varied features leading to a total of 20 experiments (except for filtered configuration which has 10% outliers removed from training set). This is for ensuring that the performance is varied solely due to preprocessing method and/or choice of machine learning models. The performance is analysed using Receiver Operating Characteristic (ROC) curve and Area Under Curve (AUC) which are well documented in the literature (Davis & Goadrich, 2006; Fawcett, 2006; Silva Araújo, Guimarães, de Campos Souza, Silva Rezende, & Souza Araújo, 2019). Cohen's kappa index (Cohen, 1960) is used to assess the inter-algorithm agreement.

### 3.7. Software used

The sampling ready geospatial feature dataset was geo-processed and exported using the JavaScript API of Google Earth Engine (Gorelick et al., 2017). The machine learning algorithms were trained and tested using the *numpy* (Oliphant, 2006), *scipy* (Jones, Oliphant, & Peterson, 2014) and *sklearn* (Pedregosa et al., 2011) packages of the Python programming language (Van Rossum & Drake, 2003). All visualizations were created using *rgdal* (Bivand et al., 2015), *raster* (Hijmans & van Etten, 2012), *sp* (Bivand, Pebesma, Gómez-Rubio, & Pebesma, 2008; Pebesma & Bivand, 2005) and *ggplot2* (Wickham, 2016) packages of the R language (Team, R. C., 2013).

## 4. Results and discussion

### 4.1. Geospatial feature profile and landslide risk assessment

As mentioned previously, three of the four features are continuous and numeric (elevation, NDVI and TWI) and one feature is nominal categorical (lithology) shown in Figure 3. The VIF of the numeric geospatial features, elevation, NDVI and TWI are 3.53, 2.87 and 4.57

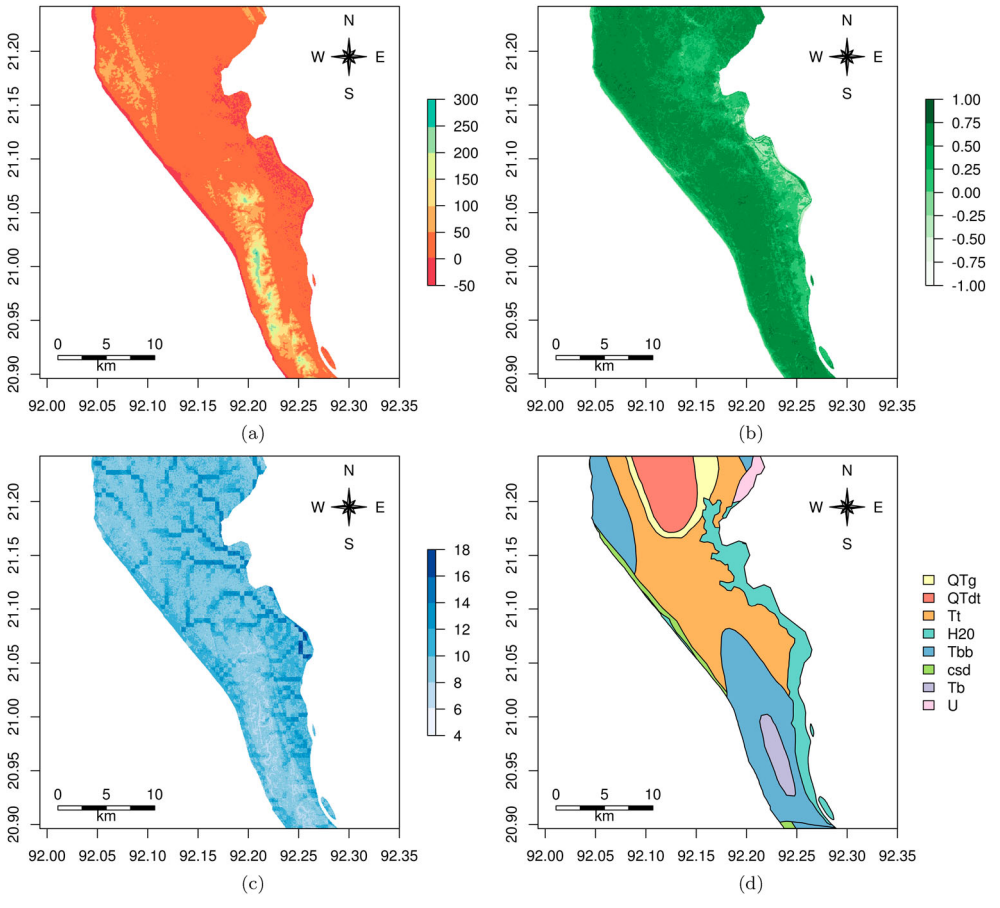



**Figure 3.** Geospatial feature profile of study area: (a) elevation, (b) NDVI, (c) TWI, (d) lithology.

respectively. The graphical representation of elevation data of the study area is shown in Figure 3(a). Considering the figure, it is evident that majority of the southeastern border region of Bangladesh has elevation between 0 m and 50 m. Additionally, it can also be observed that the highest elevation intervals ( 200–250 m and 250–300 m) are in the southern part of the study area (east of Bardeil (lime green region in Figure 3 a). The lowest elevation ranges between −50 and 0 m which is observed to be the Bay of Bengal shoreline (western side of study area) and the drainage basin of the Naf river (eastern side of study area). The northernmost and largest camp, Kutupalong, contains 95.77% of total number of infrastructure ( 16,928 of 17,675) and contains 99.68% of infrastructures with the presence of landslide risk (4677 of 4692). Figure 4 shows the geospatial feature profile of Kutupalong megacamp along with locations of infrastructure with landslide risk absent or present. Figure 4(a,b) shows that most camp infrastructure are located in low elevation regions.

Figure 3(b) shows that the area has high NDVI in most regions. This is due to the inclusion of the Teknaf Game Reserve in the analysis of study area. However, Figures 1 and 4(b) show that NDVI value is low at certain locations of camp infrastructure due to clearing of the forest and other vegetation for placement of said infrastructure. The basin of the Naf River also has

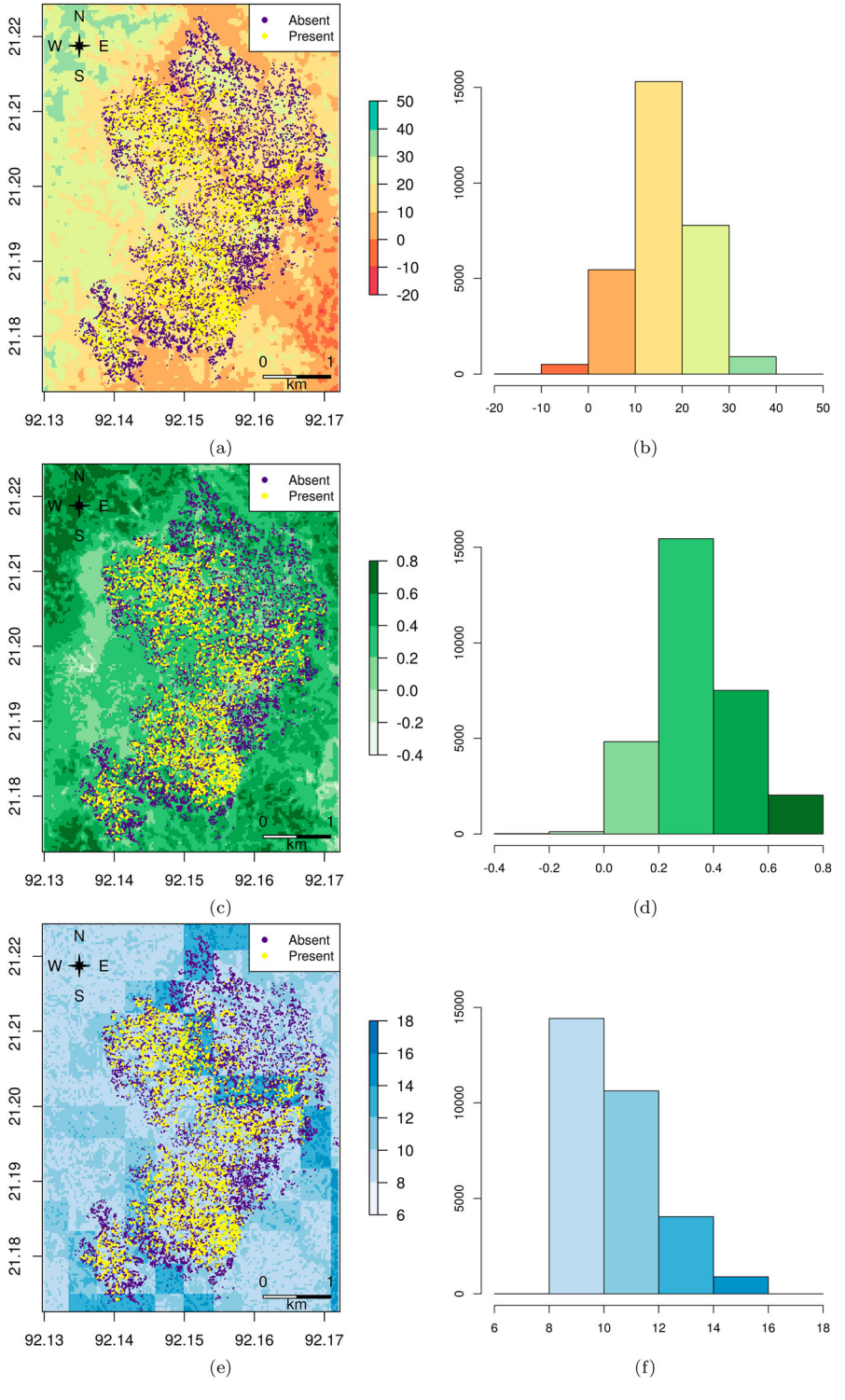



**Figure 4.** Kutupalong megacamp: (a) elevation; (c) NDVI; (e) TWI; (b), (d) and (f) – histogram of features. Absent – Landslide Risk Absent; Present – Landslide Risk Present.

low NDVI values, showing no vegetative grounds as it is a water body. Figure 4(c) shows that regions with lowest NDVI in Kutupalong are prone to landslide risk whereas regions at edges of camp (closer to vegetation) have absence of landslide risk. Vegetation root systems reduce landslide risk by increasing stability of the soil layer (Anderson & Holcombe, 2013; Dolidon, Hofer, Jansky, & Sidle, 2009; Renaud, Sudmeier-Rieux, & Estrella, 2013) suggesting that forest expansion is possibly the simplest and most effective means of landslide risk reduction. While there are other means of landslide risk reductions, such as laying concrete slabs along the slope of a hill or mountain, it can be labour intensive and financially exhausting for the managing authority. Though alternative methods have proven to work, they are not feasible in the long run. However, placement of camp infrastructure requires cleared (un-forested) land, contributing to increased landslide risk factor. A planned construction of refugee camps in perspective of landslide risk areas, which would retain enough number of trees or other vegetation, would be one effective way to reduce the impact of landslides. Figure 3(c) shows that the feature TWI is an indicator of the spatial drainage displacement (usually due to depressions in the terrain). Lower values of TWI indicate less water flowing over the specific cell (30 m × 30 m) whereas higher values indicate higher magnitude of water flow over the respective cell. As Figure 3(c) shows, water flows in networks (tree like structures) where the root of the tree has the highest TWI. The interval with the highest frequency is from 8 to 10, followed by 10 to 12. This is because more pixels of interval 8–10 are carrying the same amount of water as less pixels of interval 10–12, explaining the higher TWI values. The highest TWI values ( 16–18) are observed in the Naf river basin (eastern side of study area) because the water accumulated from the mountainous regions flows through a small area, thereby giving high TWI values. For Kutupalong, higher density of infrastructure with landslide risk can be observed in regions neighbouring large drainage depressions as shown in Figure 4(e).

The abbreviated forms of lithology types as well as number of corresponding camps having absence and presence of landslide risk are shown in Table 3. The risk ratio denotes the ratio of number of infrastructure with landslide risk presence to the number of infrastructure with landslide risk absence.

Figure 3(d) shows the lithology of the study area which mainly contains eight types of lithological categories. However, the camp block infrastructures are on six of eight types, as shown in Table 3. Figure 5 shows that Kutupalong megacamp has four lithology types, i.e. *QTg*, *QTdt*, *Tt* and *H20*. Even though *QTg* has more infrastructure with landslide risk higher than *QTdt*. *QTdt* has higher risk ratio which means 54.38% of infrastructure falling within *QTdt* lithology type were at risk of landslides (highest among all lithology types). On the contrary, lithology type *csd* has the lowest risk ratio (0%). However, 18 (0.001% of total 17,675 infrastructures) camps that fall within *csd* regions are situated closely along water bodies (such as Bay of Bengal). *Tbb* with second lowest risk ratio (1.96%) includes 0.58% of 17,675 infrastructures. It is observed that comparatively less infrastructure have been placed in regions with lithology types with lower landslide risk. The lithological characteristic needs to be taken under crucial consideration when constructing infrastructure for refugees.

Table 4 shows that latrines have the highest landslide risk ratio at 40.23%, followed by hand pump tubewells at 37.49%. This is because, even though general infrastructure, health facilities and learning centres are important facilities for refugees, hand pump tube wells and latrines are integral part of life and therefore required by the entirety of the Rohingya population in the camps. Most importantly, latrines and tube wells are

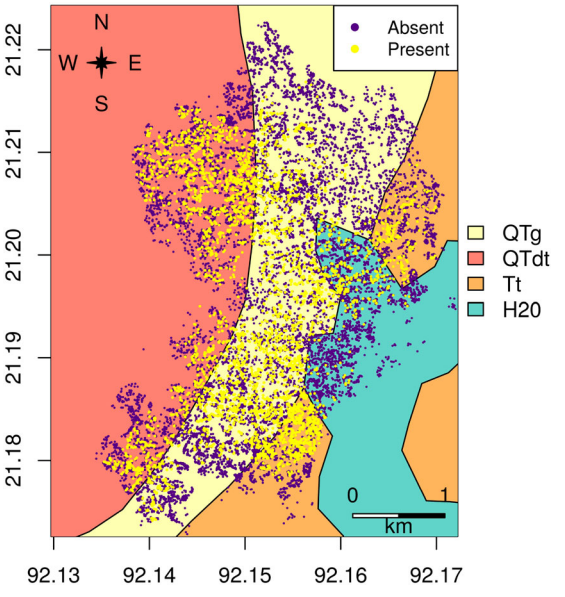



**Figure 5.** The lithology categories of Kutupalong megacamp and spatial distribution visualization.

essential due to its inseparable relation to sanitary health risk and access to safe water (for drinking and household use) for the refugee population. Thus much more emphasis is given on the quantity or number of these necessities rather than on stability/landslide risk. Considering the current recorded Rohingya refugee population of 909,919 (UNHCR, 2019), the number of hand pump tube wells and latrines are 5945 and 9348, respectively (from Table 2). On average, about 153.06 and 97.34 people are dependent on a single hand pump tube well and latrine, respectively. This shows that an important category of infrastructure for refugees are at-risk. Then again, the construction of such necessary infrastructure in landslide safe area would require many refugees to walk longer distances. This may instigate the refugees living further away from the facilities to shift to open dumping and using unsafe water, thereby causing greater sanitary, health and solid waste management related problems among the already vulnerable population.

### *4.2. Performance evaluation*

The accuracy is the percentage of correct predictions made on testing data. The bootstrap is used where samples are chosen with replacement. The training and testing of each model have been performed 1000 times. Table 5 shows the statistics of accuracy and

**Table 4.** Landslide risk summary of infrastructure types.

| Type | Landslide risk absent | Landslide risk present | Risk ratio |
|---|---|---|---|
| General infrastructure | 1183 | 182 | 0.1538 |
| Health facilities | 140 | 28 | 0.2000 |
| Learning centres | 640 | 179 | 0.2780 |
| Hand pump tube wells | 4324 | 1621 | 0.3749 |
| Latrines | 6666 | 2682 | 0.4023 |
| Total | 12,953 | 4692 | 0.3545 |

AUC of the algorithms on the five different versions of the dataset. The table shows that performance remains unaffected by preprocessing methods, i.e. algorithms have very similar (with negligible difference) performance on all 5 preprocessing varied versions of features. However, a general trend can be observed for the performance of the algorithms. LR consistently has worst performance both in terms of Accuracy and AUC. Performances of MLP and GBT are comparable and similar, although being better than that of LR. RF achieves best performance in terms of mean, standard deviation, upper and lower bounds of 95% confidence intervals of accuracy and AUC. It is quite evident that AUC of different algorithms differs in higher magnitude than test accuracy. Even though difference in test accuracy of lowest performing configuration (LR – Filtered – 73.65%) and highest performing configuration (RF – Filtered – 76.46%) is less than 3%, the difference in AUC of lowest performing configuration (LR – Normalized – 0.629) and highest performing configuration (RF – Min–Max-Scaled – 0.7622) is approximately 14% indicating that RF gives more reliable results. It is also evident that the performance of configurations with relatively similar test accuracies can be differentiated using AUC values. Each algorithm is trained on a specific preprocessing varied feature set for 1000 times where the resampling for training and testing sample set creation is achieved using the bootstrap. The mean and SD are calculated from the 1000 training-testing iterations for each algorithm on specific preprocessing varied features shows that statistical fluctuations are observed for the performance metrics.

Figure 6 shows the ROC curve of the algorithms on the un-scaled features. The Receiver Operation Characteristic (ROC) curve shows how well the configurations are able to differentiate between the two classes during testing where the False Positive Rate (FPR) and True Positive Rate (TPR) are plotted on the *x* and *y* axes respectively. The higher the

**Table 5.** Accuracy and AUC statistics of algorithms evaluated on different configurations of feature preprocessing.

| | | Accuracy | | | | AUC | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 95% Confidence interval | | | | 95% Confidence interval | |
| Algorithm | Preprocessing | Mean | SD | Lower | Upper | Mean | SD | Lower | Upper |
| LR | Un-scaled | 73.77 | 0.53 | 73.01 | 74.68 | 0.6733 | 0.008 | 0.6594 | 0.6878 |
| | Min–Max-Scaled | 73.84 | 0.53 | 73.09 | 74.8 | 0.6732 | 0.0075 | 0.6595 | 0.6879 |
| | Standardized | 73.8 | 0.53 | 73.07 | 74.75 | 0.673 | 0.01 | 0.6594 | 0.6878 |
| | Normalized | 73.73 | 0.57 | 72.99 | 74.64 | 0.629 | 0.008 | 0.6093 | 0.6368 |
| | Filtered | 73.65 | 0.61 | 73.1 | 74.61 | 0.636 | 0.0085 | 0.6071 | 0.6587 |
| MLP | Un-scaled | 74.28 | 0.53 | 73.49 | 75.24 | 0.6923 | 0.0078 | 0.6774 | 0.7084 |
| | Min–Max-Scaled | 74.46 | 0.57 | 73.7 | 75.55 | 0.6976 | 0.008 | 0.6831 | 0.711 |
| | Standardized | 74.48 | 0.66 | 73.67 | 75.68 | 0.7058 | 0.0076 | 0.6882 | 0.7182 |
| | Normalized | 74.14 | 0.51 | 73.47 | 75.08 | 0.685 | 0.0082 | 0.6732 | 0.6999 |
| | Filtered | 74.61 | 0.52 | 73.56 | 75.03 | 0.682 | 0.0083 | 0.6724 | 0.701 |
| GBT | Un-scaled | 74.68 | 0.61 | 74.09 | 75.85 | 0.7089 | 0.0068 | 0.6988 | 0.7216 |
| | Min–Max-Scaled | 74.66 | 0.62 | 74.01 | 75.87 | 0.709 | 0.0076 | 0.6986 | 0.7218 |
| | Standardized | 74.68 | 0.61 | 74.09 | 75.85 | 0.7089 | 0.0074 | 0.6988 | 0.7216 |
| | Normalized | 74.66 | 0.58 | 73.93 | 75.71 | 0.7107 | 0.0078 | 0.6955 | 0.7217 |
| | Filtered | 74.09 | 0.56 | 73.88 | 75.67 | 0.7077 | 0.0076 | 0.6889 | 0.7216 |
| RF | Un-scaled | 76.19 | 0.59 | 75.36 | 77.11 | 0.7602 | 0.0068 | 0.7468 | 0.7687 |
| | Min–Max-Scaled | 76.31 | 0.37 | 75.62 | 76.83 | 0.7622 | 0.0063 | 0.7541 | 0.7763 |
| | Standardized | 76.22 | 0.53 | 75.11 | 76.8 | 0.7612 | 0.007 | 0.7488 | 0.7798 |
| | Normalized | 76.28 | 0.58 | 75.31 | 77.25 | 0.7592 | 0.0076 | 0.7453 | 0.7765 |
| | Filtered | 76.46 | 0.59 | 75.31 | 77.11 | 0.7611 | 0.0081 | 0.7463 | 0.7765 |

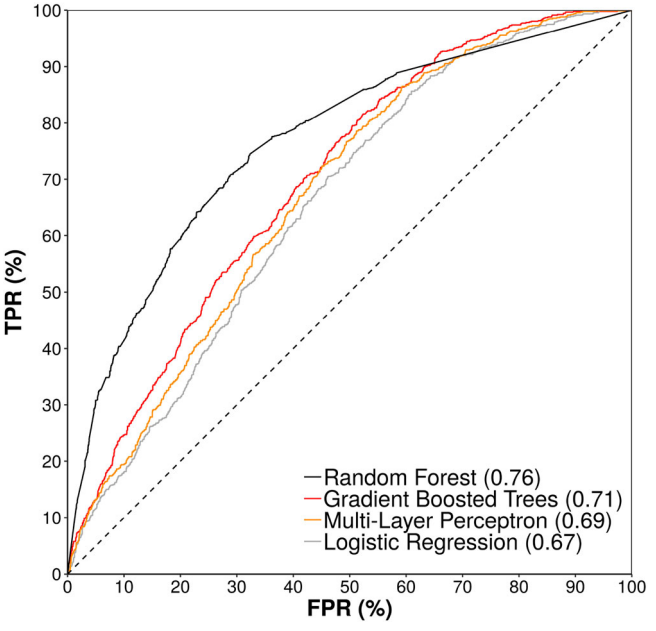



**Figure 6.** ROC curve of machine learning algorithms on un-scaled features.

area under the ROC curve known as Area Under Curve (AUC) the better a configuration is at differentiating samples correctly. Considering Figure 6, the farther away a curve is from the dashed diagonal the higher its AUC. The LR configuration is closest to the diagonal whereas the RF configuration are the farthest away. MLP and GBT configurations lie between the LR and RF curves.

In the case of landslide prediction, missed alarms are more dangerous than false alarms. Thus more emphasis is put on lower number of missed alarms and threshold is chosen accordingly, specially since there is a class imbalance (number of infrastructure at risk are lower than number non-risk infrastructure), which leads to the selection of lower values of optimal threshold. Figure 7 shows the confusion matrices of the four algorithms. The threshold values for LR, MLP, GBT and RF are 0.3, 0.3, 0.35 and 0.4 respectively. It is evident that the algorithms are very efficient at detecting negative labels.

The kappa statistic shown in Figure 8 is derived from the testing on un-scaled features and reveals that MLP and GBT have highest agreement (0.7). However, all algorithms have low agreement with RF. Considering the fact that, RF has highest accuracy and AUC, it is observable that the agreement of RF with other algorithm is low because other algorithms are making errors which only RF is classifying correctly.

## 5. Conclusion

The geospatial feature profile of the study area and infrastructure locations provide insight into the factors affecting landslides. A major proportion of the study area has elevation between 0 m and 50 m. The prevalent interval of NDVI in the study area is 0.5–0.75 indicating the presence of abundant dense live vegetation. However, the NDVI values are low at locations of camp infrastructure due to deforestation and/or removal of vegetation.

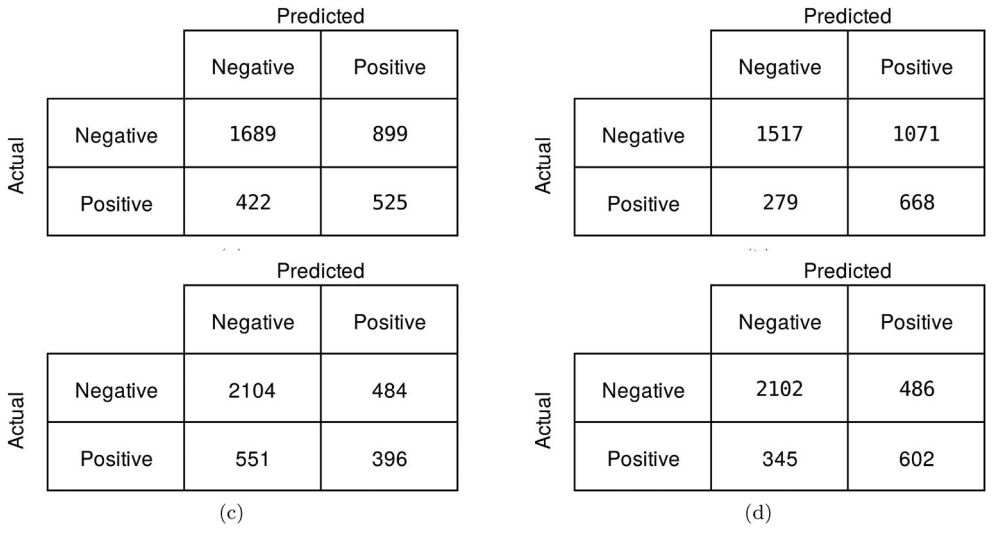



**Figure 7.** Confusion matrices for (a) LR, (b) MLP, (c) GBT and (d) RF.

The Naf river basin, adjacent to the Kutupalong Mega Camp (northern most camp), has highest TWI values ( 16–18). The number of infrastructure placed on low risk lithology type is lower than in high risk lithology type, i.e. 54.38% of infrastructures falling within *Qtdt* lithology type were at risk of landslides. Latrines have the highest landslide risk ratio, i.e. 40.23% are prone to landslides, followed by hand pump tubewells, 37.49% of which are prone to landslides. RF algorithm has better performance in terms of Accuracy and AUC than all other algorithms. The agreement of RF with other algorithms is also low. Results show that RF achieves accuracy of 76.19% and AUC of 0.76 on un-scaled features. As of 2019, US$920.5 million has been requested by the Joint Response Plan which encompasses sanitation and shelter aid for the Rohingya refugees (OCHA, 2019). Emphasis and aid should be directed to creation and curation of more detailed datasets, i.e. of better

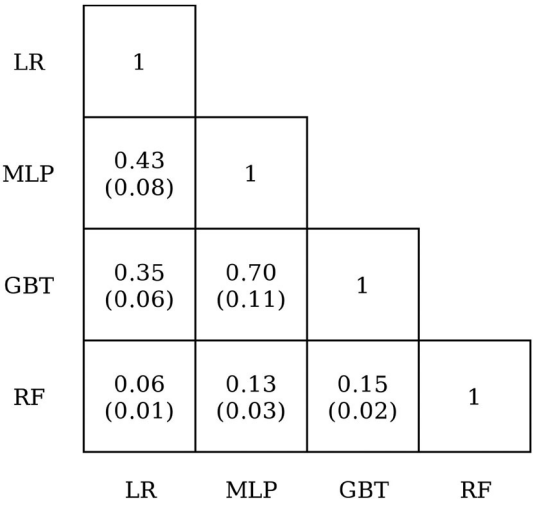



**Figure 8.** $\kappa$ statistics of algorithms showing inter-rater agreement.

spatial resolution, of the camp locations, which will contribute to higher accuracy of machine learning algorithms for predicting if potential future locations for camps are at high risk of landslides. For example, Light Detection and Ranging (LiDAR)-based datasets can provide much more detailed features, increasing accuracy and reliability of the machine learning algorithms applied.

## Disclosure statement

No potential conflict of interest was reported by the authors.

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